

Supersonic Rocket Protuberance Shape Optimization Using Response Surface and Adjoint Methods

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I. Introduction

Asymmetric protuberances along slender supersonic bodies can lead to aerodynamic instabilities due to the generation of lateral forces [1]. Consequently, the overall aerodynamic performance of the vehicle may be significantly affected. Such protuberances may correspond to camera fairings, antenna mounts, fluid pipes, reaction control systems, among others [2].

The influence of protuberance location on side-force generation was previously studied by [3]. Their results showed that the largest side-force coefficients are generated when the protuberance is located in the upper section of the rocket with an azimuthal position of 45° . In addition, [1] investigated the effects of protuberance size for a fixed longitudinal position, concluding that the protuberance height has a greater influence on the side-force coefficient than its width. Similarly, [4] analyzed the effects of the protuberance longitudinal position on the roll coefficient, revealing that the highest roll coefficients occur when the protuberance is located near the middle section of the rocket.

Although previous studies have identified the primary geometric and positional parameters influencing aerodynamic coefficients, little attention has been devoted to the shape optimization of external protuberances to minimize their aerodynamic impact in supersonic regimes. Furthermore, high-fidelity aerodynamic optimization procedures typically demand considerable computational resources, often limiting their implementation in academic and low budget research environments.

To address these limitations, this study proposes a low-cost, high-reliability optimization framework designed to deliver accurate results with reduced computational turnaround times. This framework is applied to optimize a camera fairing for the *Cattleya* rocket—developed by the Volta Rocketry team for the International Rocket Engineering Competition (IREC) 2026—with the specific objective of minimizing the side-force coefficient (C_Y) and drag coefficient (C_D) to enhance vehicle stability. Two distinct optimization approaches are implemented and compared: a Response Surface Methodology (RSM) based on a Central Composite Design (CCD) executed in Ansys Fluent, and an adjoint-based optimization procedure performed using SU2.

The first section of this study presents the methodology, including the numerical setup, mesh independence study, validation procedure, geometry definition, flow conditions, and optimization strategies. The second section presents the preliminary results obtained from the validation and optimization processes applied to the current rocket configuration. Finally, the expected outcomes and conclusions are discussed.

II. Methodology

Aiming to reduce computational cost while maintaining adequate numerical accuracy, the following methodology was implemented. First, a reference case presenting similar aerodynamic characteristics and flow conditions to those of the present study was selected for the mesh independence and validation procedures. Different mesh resolutions and solver configurations were then evaluated to identify an appropriate balance between computational cost and numerical accuracy.

After obtaining acceptable validation results, the geometry parametrization procedure is implemented using Hermite cubic interpolators. To verify that the selected design variables adequately influenced the aerodynamic response, the linearized perturbation velocity-potential equation is first employed as a low-fidelity approximation of the problem. A preliminary parametric screening is subsequently conducted over the design space using this approach.

The low-fidelity results are then compared against 2D CFD simulations, after which the parametric screening procedure is repeated. Once the selected parameters demonstrated sufficient influence over the main flow characteristics,

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full 3D CFD simulations are performed. The simulation matrix is established using a Central Composite Design (CCD). The resulting dataset was subsequently fitted using two surrogate modeling approaches: a second-order polynomial regression model and a Kriging surrogate model. For both approaches, the optimization objective consisted of minimizing the side-force C_Y and drag C_D coefficients.

In parallel, an adjoint-based optimization procedure is implemented in SU2, also targeting the minimization of the side-force coefficient. Consequently, three optimization strategies are evaluated: second-order polynomial regression, Kriging surrogate modeling, and adjoint optimization. The optimal geometries predicted by each method are subsequently validated using CFD simulations. Finally, the optimized configurations are compared against the reference geometry, and an overall assessment of the optimization methodologies was performed in order to identify the most effective approach.

A summary of the methodology is presented in Figure 1.

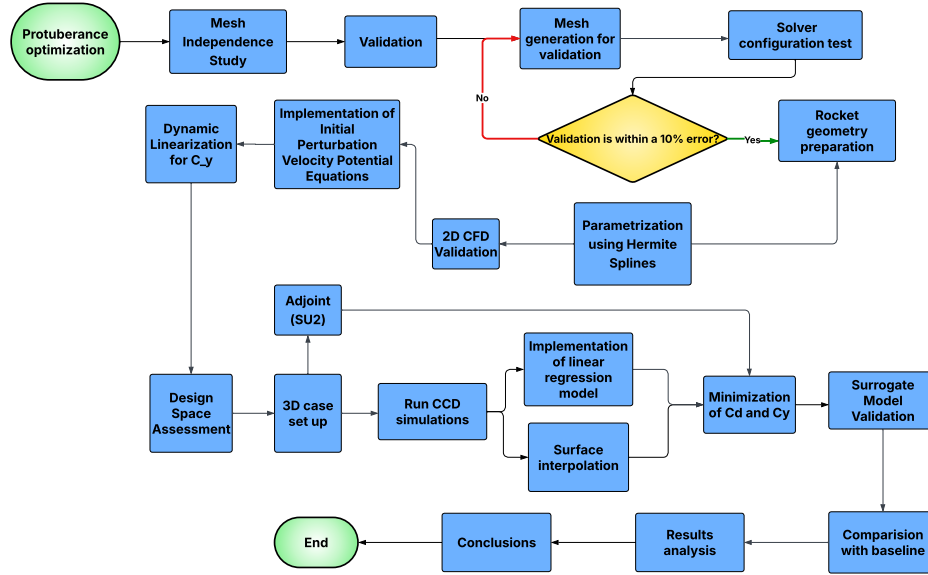


Fig. 1 General Workflow.

A. Computational Methods

For the numerical set up, the results reported by [5] were considered. In their study, the SA, SA-R, $\kappa - \omega$ SST, and SST 2003 turbulence models were compared, showing that the SST 2003 and SA-R models provided the best agreement with experimental data, it was also concluded that the SA model without correction achieves a sufficiently good approximation to be considered appropriate. Based on these findings, the SA turbulence model was selected for the present work due to its lower computational cost, which is one of the main objectives of this study. Additionally, the AUSM flux scheme was chosen because of its suitability for compressible aerodynamic flows. Both first- and second-order discretization schemes were evaluated in order to establish a trade-off between accuracy and computational cost. Air was modeled as an ideal gas, while Sutherland's law was used to estimate the dynamic viscosity.

B. Geometry and Parametrization

1. Geometry of Interest

The geometry of interest consists of a camera fairing mounted on the external surface of the Cattleya rocket, developed by the Volta Rocketry team for IREC 2026. The rocket body is a cylindrical tube of diameter $D = 174$ mm and total length $L = 3240$ mm, featuring a Von Kármán ogive nose cone and trapezoidal fins at the base. The camera fairing is located on the upper section of the rocket body at an azimuthal position of 45° , consistent with the configuration that produces the largest side-force coefficients as reported by [1]. The fairing geometry is defined by its height $h = 22$ mm, base length $l = 87$ mm, and frontal width $w = 51$ mm. These dimensions are constrained by the

physical requirements of the onboard camera system. The side profile of the fairing, which constitutes the primary aerodynamic surface, is parameterized using Hermite cubic interpolators as described in II.B.2.

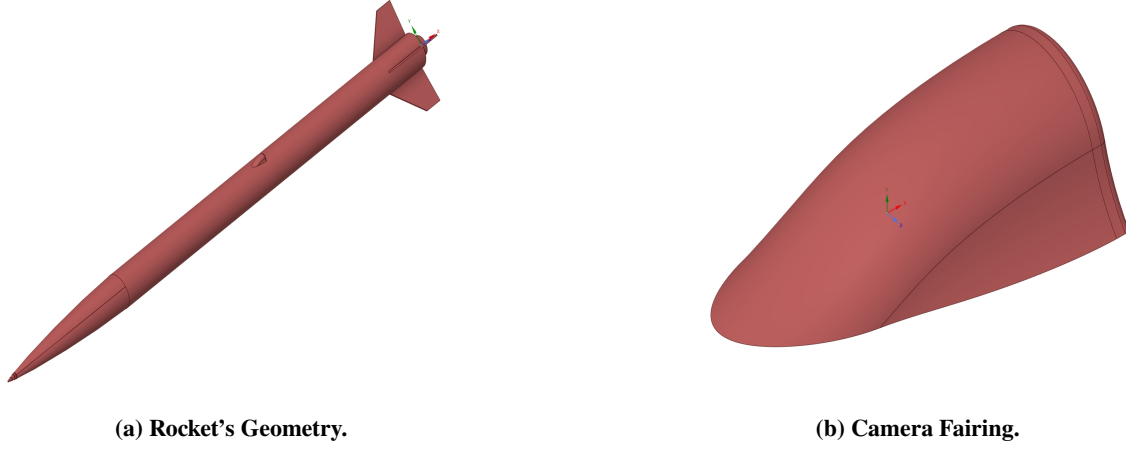


Fig. 2 Rocket geometry and camera fairing geometries.

2. Hermite Cubic Parametrization

The side profile of the camera fairing was parameterized using Hermite cubic interpolators. The final geometry is governed by the tangent vectors at the start and end points of the curve; therefore, these tangent vectors were selected as design parameters.

Only the side profile was parameterized, since [1] demonstrated that the frontal width of the protuberance has a significantly smaller influence on vortex intensity compared to the side profile geometry. Once the design parameters were defined, the experimental design was established using a Central Composite Design (CCD).

The Hermite cubic polynomial is given by Equation 1, while its corresponding derivative is presented in Equation 2.

$$X(t) = a_3t^3 + a_2t^2 + a_1t + a_0 \quad (1)$$

$$X'(t) = 3a_3t^2 + 2a_2t + a_1 \quad (2)$$

Using the boundary conditions, namely the coordinates of the start and end points together with their derivatives, the coefficients a_i can be expressed as:

$$a_0 = p_0 \quad (3)$$

$$a_1 = p'_0 \quad (4)$$

$$a_2 = -3p_0 - 2p'_0 + 3p_1 - p'_1 \quad (5)$$

$$a_3 = 2p_0 + p'_0 - 2p_1 + p'_1 \quad (6)$$

where $X(0) = p_0$ and $X(1) = p_1$. Equations 3–6 can be rewritten in matrix form as

$$X(t) = \begin{bmatrix} t^3 & t^2 & t & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & -2 & 1 \\ -3 & -2 & 3 & -1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} p_0 & p'_0 & p_1 & p'_1 \end{bmatrix} \quad (7)$$

Equation 7 allows the computation of the x and y coordinates defining the fairing profile. For the present application, the tangent lines at both the start and end points must remain parallel to the rocket surface. Therefore,

$$\left. \frac{dy}{dx} \right|_{x=0} = 0, \quad \left. \frac{dy}{dx} \right|_{x=1} = 0 \quad (8)$$

Additionally, the y-coordinates of both points are fixed by the rocket surface and the camera height. Assuming a fixed axial position for the camera, only three independent parameters are required to define the Hermite curve: x_0, x'_0, x'_1

Furthermore, defining $\Delta x = x_1 - x_0$, the parameterization can be reduced to the two nondimensional variables given in Equation 9:

$$\zeta_1 = \frac{x'_0}{\Delta x}, \quad \zeta_2 = \frac{x'_1}{\Delta x} \quad (9)$$

Figure 3 presents an example of a camera fairing profile generated using the Hermite curve parameterization.

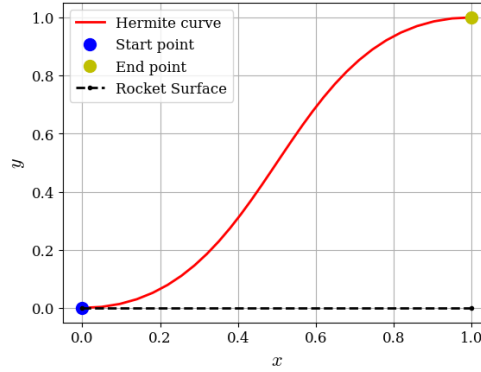


Fig. 3 Hermite profile example for the camera fairing.

C. Mesh Generation and Grid Independence

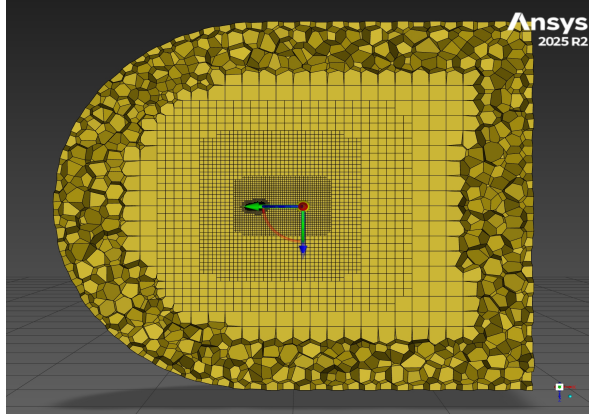
1. Mesh Strategie

The computational mesh is generated using Ansys Fluent Meshing. The domain starts from the base of the rocket, extending downstream as a cylinder, with a length of $15L$ and a diameter of $24L$, and forwards it extends as a hemisphere with a radius of $12L$.

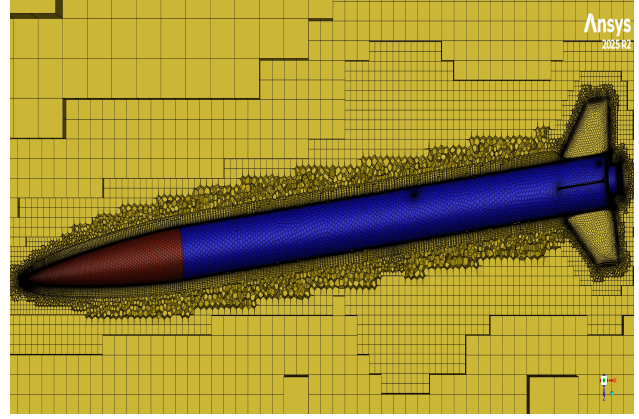
The rocket surface is divided into distinct zones, each assigned an independent sizing configuration to balance resolution and cell count. The nose cone is meshed using a curvature-based size control, ensuring adequate refinement in regions of high geometric curvature. The cylindrical body is treated with a uniform face size control. The fins are decomposed into three sub-regions: the fin tips, meshed with face size control; the root junction with the rocket body, refined using edge size control; and both the leading and trailing edges, controlled via a proximity-based sizing function. Together, these configurations yield a surface mesh generated with size functions set to Curvature & Proximity.

Following surface mesh generation, an improve surface mesh step is applied to enhance element quality prior to volume filling. The computational domain consists of a single fluid region with the rocket body treated as a dead zone enclosed within it. Within the boundary of the computational domain, there is the Farfield and the Outlet, without the use of any symmetry because the simulation of the entire domain is performed.

To control volumetric element growth from the rocket surface toward the domain boundaries, four bodies of influence (BoI) are employed. Three concentric BoIs govern the progressive coarsening of elements as distance from the surface increases. A fourth elongated BoI, oriented along the rocket axis and shaped to envelop the camera fairing region, is included to capture the vortical structures generated by the protuberance.



(a) Mesh Domain.



(b) Rocket Boundary Layer.

Fig. 4 Computational mesh domain and boundary layer refinement around the rocket.

A prism layer is then added to the rocket surface to resolve the boundary layer. The volumetric mesh is subsequently generated using a poly-hexcore fill method, producing a semi-structured mesh. Octree refinement parameters are adjusted to enforce a 2:1 transition ratio between adjacent cell sizes, and a post-processing orthogonal quality improvement step is applied to the final volume mesh.

2. Grid independence study

For the mesh independence study and the validation the rocket configurations analyzed by [1], [5], [3], and [4] is taken as reference. On section II.D it is presented the general Flow-Field conditions for the mesh independence and validation studies respectively.

Three meshes are considered: a coarse, medium and fine mesh, with 2×10^6 , 4×10^6 and 7×10^6 cells respectively. The free-stream conditions were defined as follows: $M_\infty = 1.5$, $P_\infty = 53$, kPa, $Re_L = 1.38 \times 10^7$, and an angle of attack of $\alpha = 5^\circ$. Since the geometry analyzed in the present study differs from the reference cases, all mesh refinements and sizing parameters were defined with respect to a reference length, taken as the body diameter. Therefore, differences in the overall size of the present geometry relative to the reference configurations do not affect the reliability of the mesh independence study.

D. Validation Set Up

The aerodynamic coefficients reported by [3] for C_A and C_Y were used as validation data to verify that both the mesh and the numerical set up provide reliable results. For the validation case, the same flow conditions were maintained while varying the angle of attack from 0° to 5° . A summary of the flow conditions for the mesh independence and validation is presented in Table 1. The geometry of the model used by [1], [5], [3], and [4] is shown on Figure 5 with reference dimensions.

Variable	Value
M_∞ [-]	1.5
P_0 [kPa]	196.1
T_0 [K]	294
Re_L [-]	1.38×10^7
α [°]	0, 1, 2, 3, 4, 5
γ [-]	1.4
R [J/(kgK)]	287

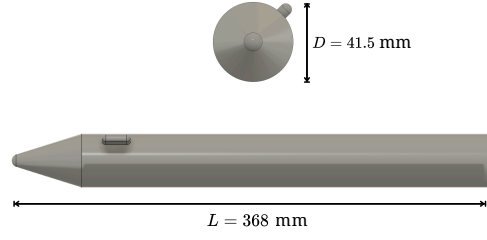


Table 1 Flow-Field properties for validation and mesh independence.

Fig. 5 Reference geometry used by [3]

E. Multi-Fidelity Screening Strategy

1. Linearized Perturbation-Velocity Potential Model

As a first approximation to the problem and to assess the adequacy of the parameterization procedure, the linearized perturbation velocity-potential equation for two-dimensional supersonic flow was considered:

$$\lambda^2 \frac{\partial^2 \phi}{\partial x^2} - \frac{\partial^2 \phi}{\partial y^2} = 0 \quad (10)$$

where

$$\lambda = \sqrt{M_\infty^2 - 1} \quad (11)$$

and ϕ represents the perturbation velocity potential. The general solution to Equation 10 can be expressed as

$$\phi = f(x - \lambda y) + g(x + \lambda y) \quad (12)$$

corresponding to the two characteristic families associated with supersonic flow. Assuming a dominant left-running Mach-wave family and neglecting reflected perturbations, the solution is simplified by setting $g = 0$.

The wall boundary condition is imposed according to

$$\frac{dy}{dx} = \frac{v'}{V_\infty + u'} \quad (13)$$

where u' and v' denote the perturbation velocity components, defined as

$$u' = \frac{\partial \phi}{\partial x} \quad (14)$$

and

$$v' = \frac{\partial \phi}{\partial y} \quad (15)$$

respectively.

This low-fidelity formulation provides a preliminary estimation of the pressure coefficient distribution along the protuberance surface while requiring minimal computational resources. Since the analysis is restricted to small angles of attack and weak perturbations, a linear relationship between the side force coefficient and angle of attack is assumed.

Owing to its low computational cost, the model is employed to perform an initial screening of the design space and to evaluate whether the selected parameterization was capable of producing meaningful variations in the aerodynamic

response. A second-order polynomial regression model is constructed from the generated dataset in order to identify potential minima within the design space. The obtained trends were subsequently compared against 2D CFD simulations, discussed in Section II.F.1.

F. CFD Configuration

1. 2D CFD set up

Prior to any CFD simulation, an analytical screening is conducted to verify that variations in the fairing geometry produce a measurable influence on the aerodynamic coefficients. Oblique shock wave theory is applied to a wedge-equivalent representation of the fairing side profile, allowing the estimation of surface pressure distributions and the resulting force coefficients as a function of the geometric design parameters ζ_1 and ζ_2 defined in II.B.2. This step confirms that the selected parameterization spans a design space with sufficient aerodynamic sensitivity before investing computational resources in higher-fidelity simulations.

Once the preliminary analysis establishes the influence of the design parameter, a two-dimensional CFD model is used as an intermediate validation step. The configuration follows the classical supersonic flow over a wedge at $M_\infty = 1.5$, for which the analytical solution via oblique shock relations serves as the reference. The computational domain is enclosed by a single pressure far-field boundary condition, specifying the free-stream Mach number, static pressure, and temperature consistent with the conditions in 1. This condition handles both supersonic inflow and outflow regions, and the wedge surface is treated as an inviscid wall. Agreement between the 2D CFD results and the analytical solution within the accepted error threshold validates the numerical setup before proceeding to three-dimensional simulations.

2. 3D CFD set up

The three-dimensional CFD simulations are performed in Ansys Fluent. Based on the findings of the mesh independence and validation study describe in II.A, the Spalart-Allmaras (SA) turbulence model is employed in conjunction with the AUSM flux scheme. Air is modeled as an ideal gas with dynamic viscosity estimated via Sutherland's law.

The pressure and temperature conditions are configured in the Far Field and Outlet, as well as the kinetic viscosity used, which is $\tilde{\nu}_{FarField} = 3\nu$ [6]. The length, diameter, and cross-sectional area of the rocket are considered as reference values for calculating the coefficients.

The solution procedure follows a two-stage convergence strategy. In the first stage, a first-order upwind discretization scheme is used to obtain a stable initial flow field, employing hybrid initialization to start the simulation. Once initial convergence is achieved, the solution is switched to a second-order upwind scheme to improve accuracy. Full Multigrid (FMG) initialization is subsequently applied to accelerate convergence of the second-order solution. Residual convergence criteria (Around $1e - 03$) and the monitoring of the side-force coefficient C_Y are used jointly to assess solution convergence.

G. Design of Experiments

To efficiently generate the response maps as functions of the design parameters, the CCD approach was employed. Within this framework, nine distinct geometries were evaluated, corresponding to four factorial points, four axial points, and five central design point. After completing the simulations, the resulting dataset was used to construct either a second-order polynomial regression model or a Kriging surrogate model. Both approaches provide a mathematical representation of the surrounding parametric space, enabling the prediction of an optimal camera fairing geometry through the minimization of an objective function based on the side-force coefficient.

H. Surrogate Modeling and Optimization

1. Second Order Polynomial Regression

Once the CCD simulation matrix is completed, a second-order polynomial regression model is constructed to approximate the aerodynamic response surface as a continuous function of the design parameters ζ_1 and ζ_2 . For two design variables, the model takes the general form:

$$\hat{f}(Z) = \hat{f}(\zeta_1, \zeta_2) = \beta_0 + \beta_1 \zeta_1 + \beta_{11} \zeta_1^2 + \beta_{22} \zeta_2^2 + \beta \zeta_1 \zeta_2 \quad (16)$$

In 16, $\hat{f}$ denotes the predicted aerodynamic coefficient and β is the vector of regression coefficients. Separate response surfaces are constructed for C_Y and C_D . The regression coefficients are determined by minimizing the sum of squared residuals between the model predictions and the CFD simulation results:

$$\beta = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad (17)$$

In 17, $\mathbf{X}$ is the design matrix assembled from the CCD sample points and $\mathbf{y}$ is the vector of CFD-computed responses.

The quality of each surrogate is assessed using the coefficient of determination R^2 , which quantifies the proportion of variance in the simulation data explained by the regression model. Values of R^2 close to unity indicate adequate fit of the response surface over the sampled design space.

Once the surrogate models for C_Y and C_D are validated, a composite objective function is defined to allow flexible weighting of the two aerodynamic responses:

$$J(Z) = w_Y \hat{C}_Y(\zeta_1, \zeta_2) + w_D \hat{C}_D(\zeta_1, \zeta_2) \quad (18)$$

In 18, w_Y and w_D

2. Kriging Surrogate Model

I. Adjoint-Based Optimization (SU2)

The parametrization procedure is coupled with a discrete adjoint optimization in SU2. This approach enables to efficiently compute the sensitivity of the aerodynamic objective function, and reducing the computational cost. The formulation is the minimization of C_Y (see equation 18), subjected to the compressible RANS equations with constraints defined by the Hermite parametrization.

$$\mathbf{R}(\mathbf{U}, Z) = 0 \quad (19)$$

Where $\mathbf{U}$ denotes the vector of conservative flow variables and $\mathbf{R}$ represent the discretized governing equations. At each iteration, the geometry is updated according to the Hermite parametrization, followed by a mesh deformation. The flow solution is obtained and the discrete adjoint equations are solved to determine the sensitivity distribution and with back propagation the gradients of the design variables are updated until convergence is reached.

III. Preliminary Results

A. Mesh independence and Validation Results

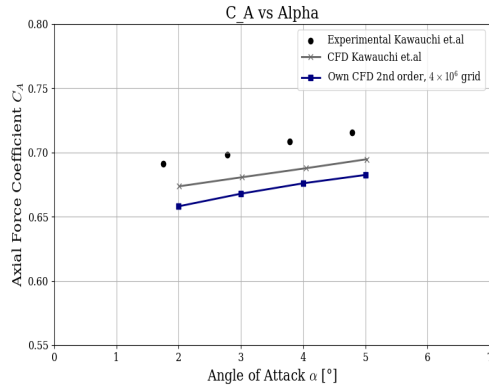
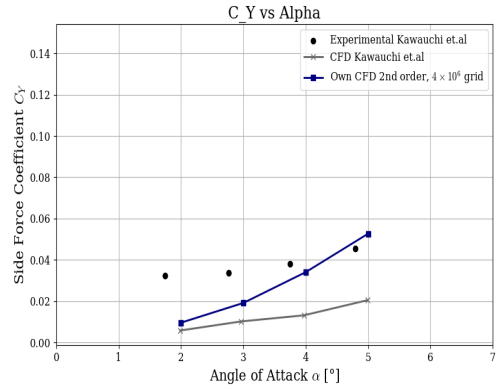
For the mesh independence study, it was found that the grid containing 2×10^6 cells exhibited convergence issues when the second-order discretization scheme was employed, particularly in the residual behavior. Consequently, a first-order discretization scheme was adopted for this mesh configuration. In contrast, the grids containing 4×10^6 and 7×10^6 cells successfully achieved convergence using the second-order discretization scheme, yielding more reliable numerical results.

A comparison between the different mesh resolutions and discretization schemes is presented in Table 2. Although the 4×10^6 and 7×10^6 cell grids with second-order discretization provided the most accurate results, the computational cost associated with the 7×10^6 cell grid was significantly higher. Since the improvement obtained with the finer grid was relatively small compared to the 4×10^6 cell grid, and considering that the present work aims to establish a computationally efficient methodology, the 4×10^6 cell grid was selected for the subsequent 3D simulations.

Table 2 Mesh independence study results.

Cells	Order	Present CFD		Experimental		Error [%]	
		C_A	C_Y	C_A	C_Y	C_A	C_Y
2×10^6	1st	0.8256	0.0383	0.7175	0.0490	15.07	21.71
4×10^6	1st	0.7738	0.0357			7.85	27.10
4×10^6	2nd	0.6825	0.0527			4.88	7.61
7×10^6	2nd	0.6899	0.0514			3.87	5.12

The validated results are shown on Figure ??, where it can be seen a good agreement between the experimental values and the results given by the CFD results, it is also noticeable the linear behavior on both the axial coefficient C_A and the side force coefficient C_Y , indicating a consistent behavior. Thus, this supports the implementation of the linearized perturbation-velocity potential model for this low angles of attack.

**(a) Axial Force Coefficient validation.****(b) Side Force Coefficient Validation.****Fig. 6 Validation results for axial and side forces coefficients.**

B. Preliminary camera fairing simulation

A preliminary simulation of the camera fairing was performed to assure that the main flow characteristics were captured. On figure 7, there is presented the pressure coefficient and surface streamlines along the rocket surface, where it can be seen an abrupt increment on the camera fairing, indicating the presence of a shock wave. At the back of the camera, a recirculation region is produced along with a separation and reattachment of the surface streamlines.

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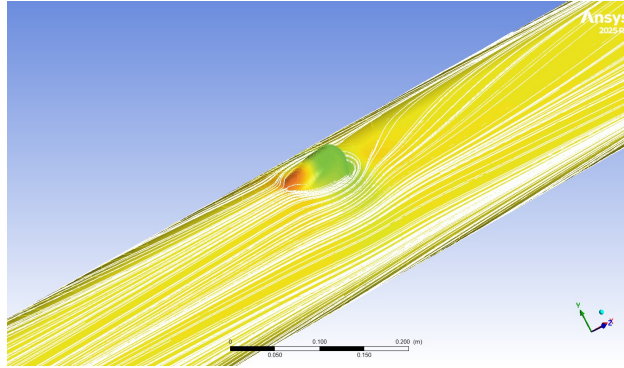


Fig. 7 Pressure Contour and Streamlines on the Rocket Surface

Asymmetric Protuberance,” *TRANSACTIONS OF THE JAPAN SOCIETY FOR AERONAUTICAL AND SPACE SCIENCES, AEROSPACE TECHNOLOGY JAPAN*, Vol. 17, No. 2, 2019, pp. 111–119. <https://doi.org/10.2322/tastj.17.111>.

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